

LECTURER: JOHN DOE

DATABASE MODELING AND DATABASE SYSTEMS

TOPIC OUTLINE

Relational Database Basics

1

Database Queries to Exactly One Table

2

Conception and Modeling of Relational Databases

3

Creating Relational Databases

4

Complex Database Queries on Multiple Tables

5

Manipulating Records in Databases

6

NoSQL Database System

7

UNIT 7

NoSQL DATABASE SYSTEMS

STUDY GOALS



- Explain the meaning and the need for NoSQL.
- Discuss the difference between constraints of relation and non-relational data models.
- Explain the differences among different available NoSQL system types.



1. List three drawbacks of the relational model.
2. What are the two modern data management needs that led to the adoption of NoSQL paradigms?
3. Many NoSQL systems are open-source. What does open-source mean?

WHY NOSQL?

- Not Only SQL
 - Release much of the SQL constraints
- hundreds of thousands of users
- huge amounts of data
- flexibility
 - schema-free or weak schema restrictions
- high availability
- scalability

NOSQL PROPERTIES

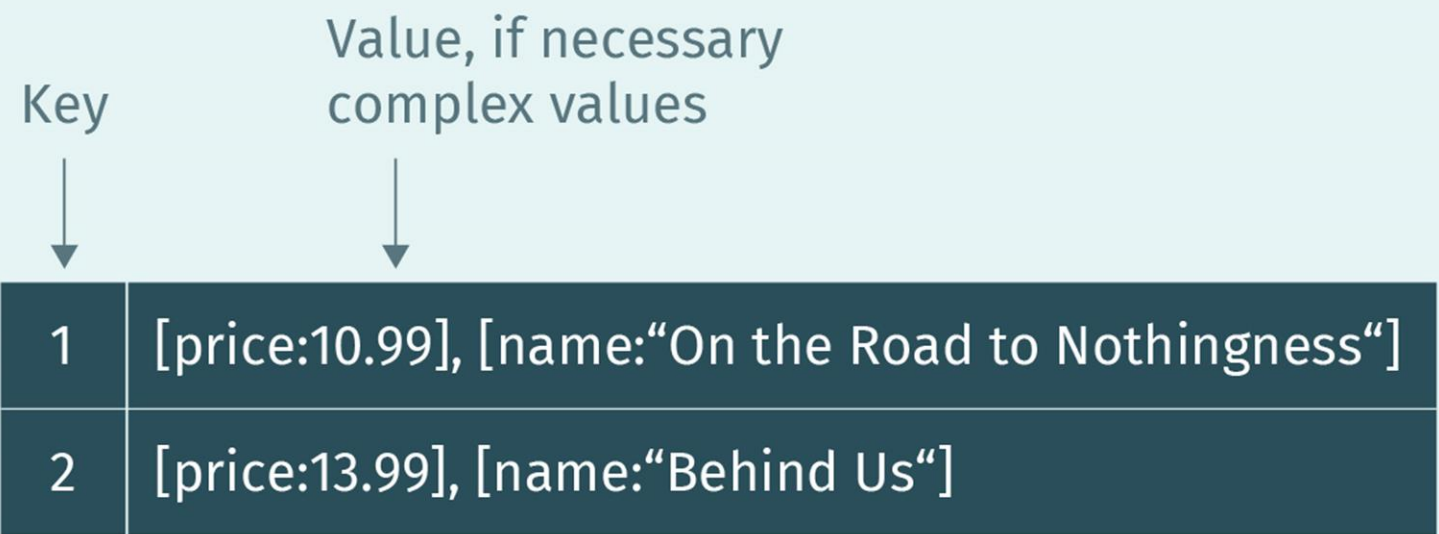
- The underlying data model is not relational.
- Distributed and horizontal scalability:
 - growing data
 - increasing users
 - simple replication
- semi- and unstructured data

BASE VS. ACID

- ACID (Relational model)
 - Atomicity
 - Consistency
 - Isolation
 - Durability

- BASE (Non-relational model)
 - Basically available
 - Soft state
 - Eventual consistency

NOSQL SYSTEM TYPES: KEY-VALUE SYSTEM



Key	Value, if necessary complex values
1	[price:10.99], [name:"On the Road to Nothingness"]
2	[price:13.99], [name:"Behind Us"]

NOSQL SYSTEM TYPES: WIDE COLUMN STORES

Physical storage of the data is column-oriented

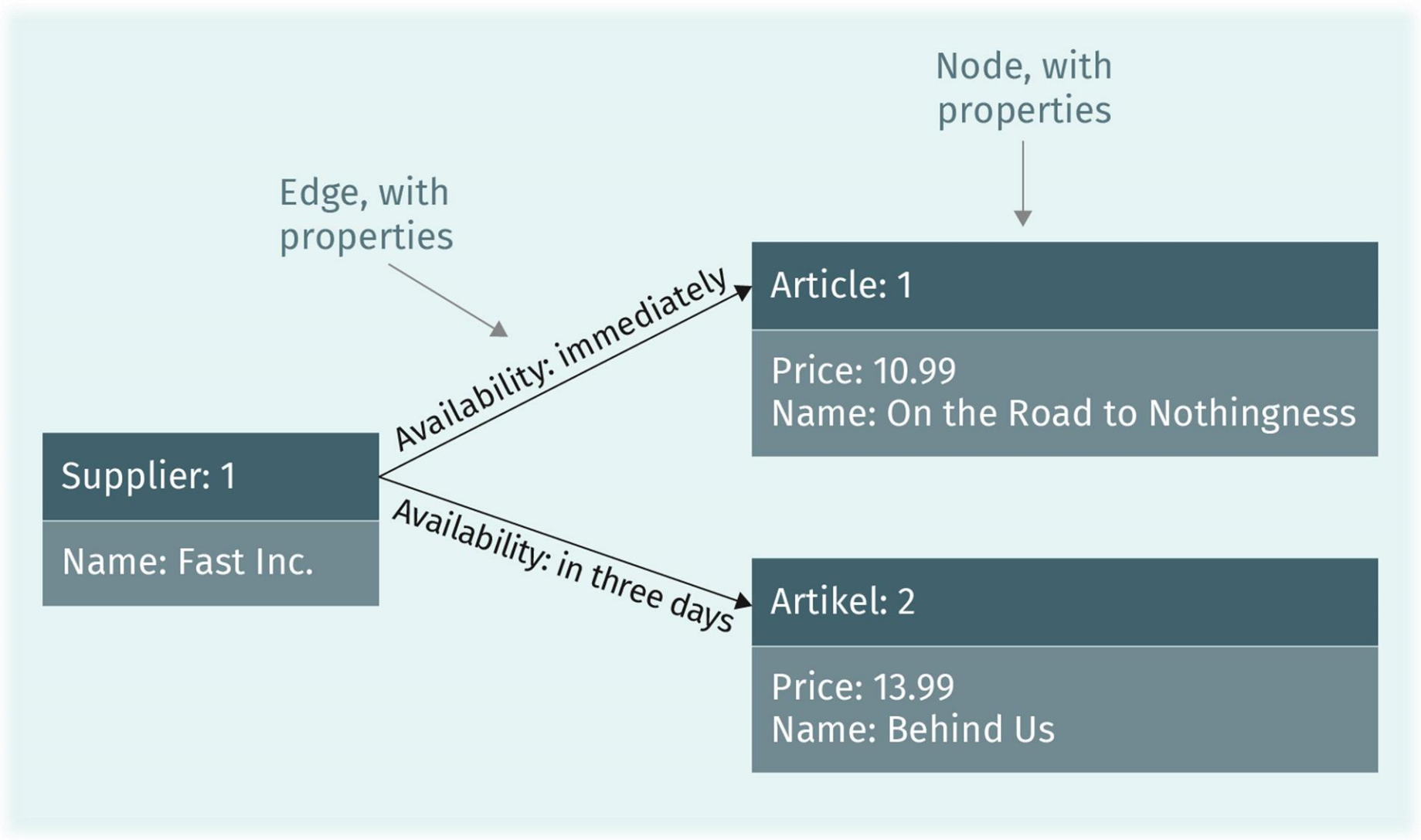


<u>ArticleID</u>	Price	Name
1	10.99	On the Road to Nothingness
2	13.99	Behind Us

NOSQL SYSTEM TYPES: DOCUMENT STORES- JSON

```
{
  "ID": "1" {
    "price" : "10.99"
    "name" : "On the Road to Nothingness"
  }
  "ID": "2" {
    "price" : "13.99"
    "name" : "Behind Us"
  }
}
```

NOSQL SYSTEM TYPES: GRAPH DATABASES



REVIEW STUDY GOALS



- Explain the meaning and the need for NoSQL.
- Discuss the difference between constraints of relation and non-relational data models.
- Explain the differences among different available NoSQL system types.

SESSION 6

TRANSFER TASK

TRANSFER TASKS

Convert the following part of a job offer letter into a JSON document:

“Dear Mr. Johns,

We are delighted to offer you the sales manager position at ABC bank, Branch 4 in Berlin, with an annual salary of €122,000 in addition to the health benefits and vacation time. The start date for this position will be on January 23. Your supervisor will be Mme. Franklin and her office number is 215.”

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your
results.

The results will be
discussed in plenary.





1. Why NoSQL databases are considered more flexible than relational databases?
 - a) NoSQL databases are schema-less.
 - b) NoSQL databases are open-source.
 - c) NoSQL databases are based on the ACID constraints.
 - d) NoSQL database schemas can be changed by the end-user.



2. What does Eventual Consistency mean with the BASE constraints?

- a) The database might or might not reach a consistent state
- b) The database will reach a consistent state eventually
- c) The database will always be inconsistent
- d) Consistency must be manually maintained by the administrator



3. Which of the following is an example of document stores:

- a) Cassandra
- b) MariaDB
- c) MongoDB
- d) Oracle

How did you
like the course?



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